Edinburgh Napier UNIVERSITY

Visualising Networks and Trees

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SET09123 – Interactive Data Visualisation

Lecture Summary

- Network and Tree Data
- Network and Tree Visualisations
- Dealing with Complexity

Network and Tree Data

How would you define Network Data?

How would you define Tree Data?

Network Data

- Network data involves both items and links between these items
  - Items get referred to as **Nodes** or **Vertices** (singular: vertex)
  - Links can also be called **Edges**

- A link will inform of a relationship between <u>two</u> nodes
  - This relationship can have different semantics: simple connection, measured distance, dependency, etc.

- Both nodes and links hold attributes
  - At the bare minimum, a link has two attributes: the two nodes it connects

Types of Links

- Directed vs. Undirected – Links can infer whether an node is visible/accessible from its neighbour or not
  - Some links can also be bi-directional

- Weighted – Links can have a weight attached to them, denoting cost, similarity, etc.

Networks vs. Graphs

- Both terms can be synonymous at times
- Graph tends to refer to the mathematical object
  - But it gets confusing for visualisation people, e.g. "*Line Graph*"
- Hence why we will use the term **Network**

Networks vs. Trees

- A **Tree** is a special type of Network typically inferring a hierarchical semantic
  - Nodes have a parent and several children

- Each node can only have exactly 1 *parent*
  - Except the **Root Node** which has 0 parent
  - Nodes with no children are referred to as **Leaf Nodes**

- Descendant nodes cannot also be ancestor nodes
  - Thus there are no cycles in a Tree

Representing Networks and Trees

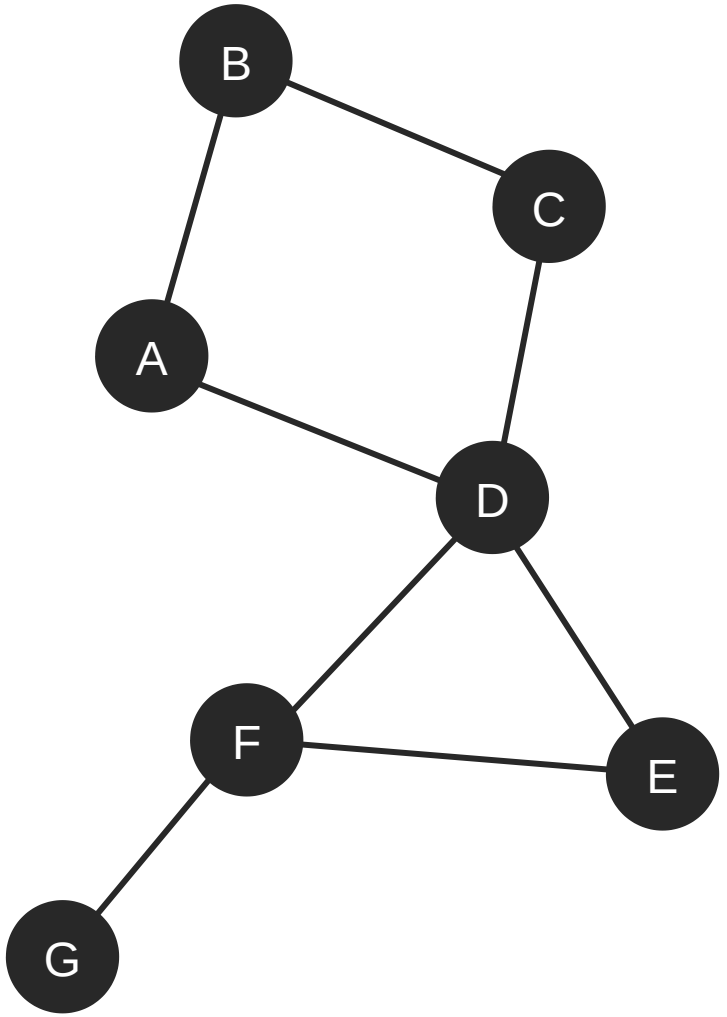
There are several strategies:

- Connecting the nodes
- Arranging nodes in matrices
- Enclosing nodes

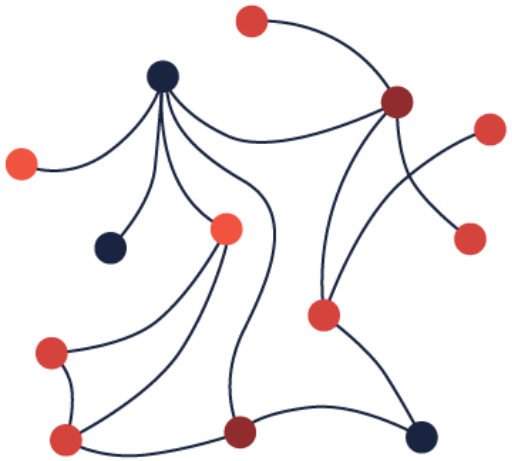
Connection

Link Marks

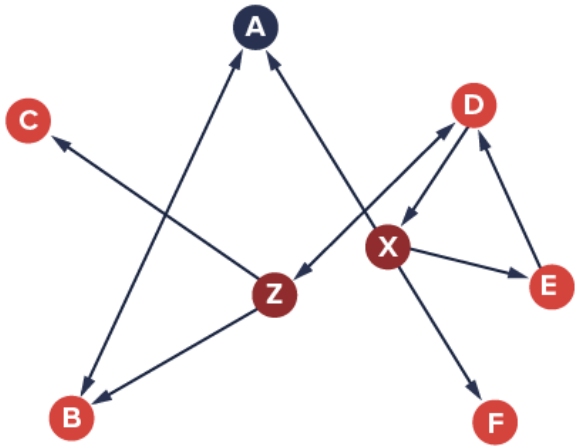
- By connecting nodes we mean actively drawing links
- Use a set of marks for the nodes
- Use another set of **line marks** for the links



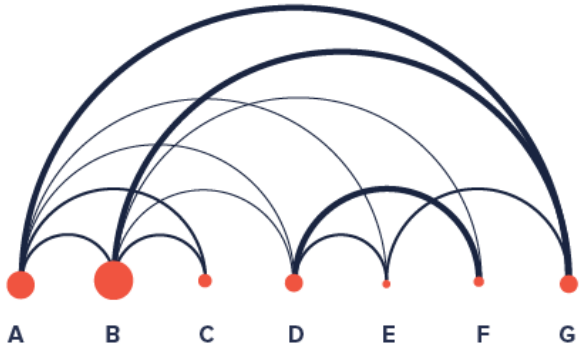
Node-Link Diagrams



Undirected



Directed



Arc Diagram

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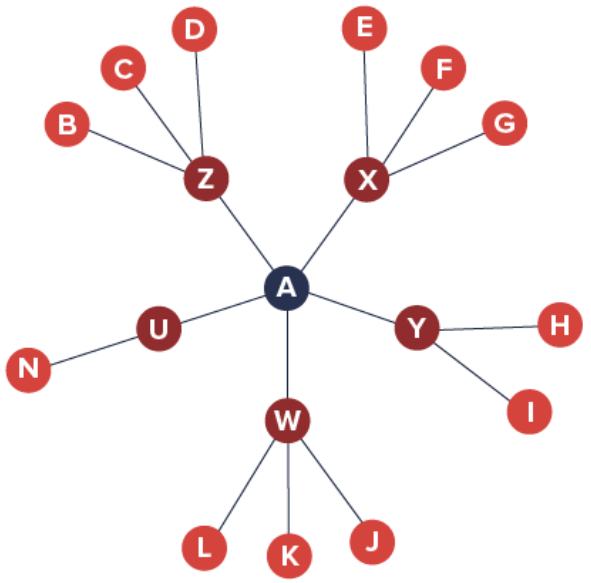
Placing Nodes

- Link placement is easy: we draw a line from one node to another
- But where do we place nodes?
- There are several considerations:
  - We should minimise links crossing for readability
  - Unless the link sizes encode data, they should have the same length
  - The network topology should be preserved, e.g., clusters or cliques

Placing Nodes

This is not much of an issue with Trees

1. Start from the root
2. Expand the children in one or more directions



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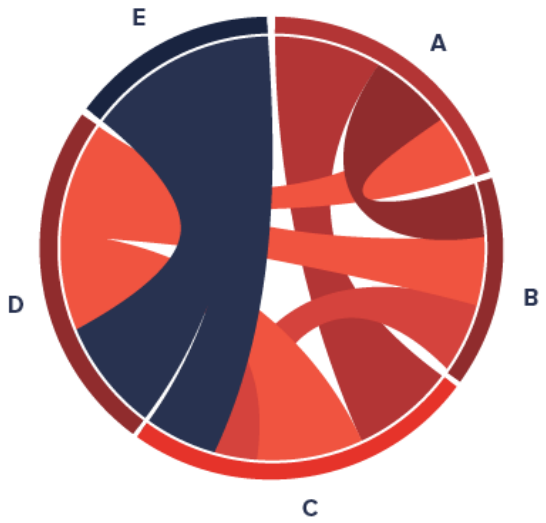
Force-Directed Layouts

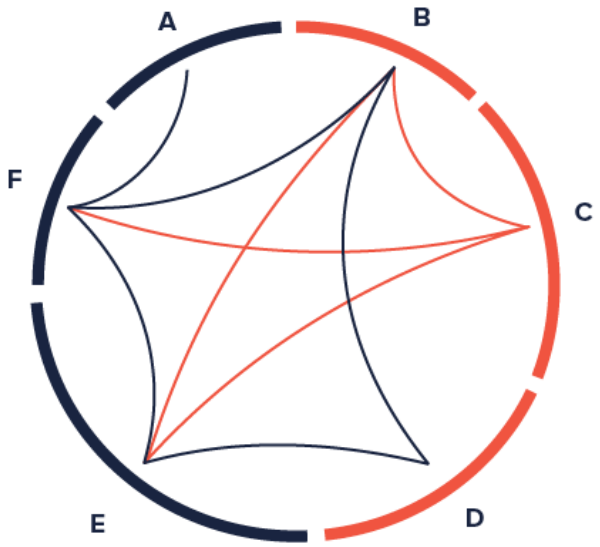
- The basic idea is to run a quick spring/gravity simulation
  - All the nodes push each other by default
  - Nodes with a connection remain attracted to each other

- The position channel becomes abstract, it's not tied to any data
  - Region becomes more important, to identify clusters
- Often based on a stochastic layout algorithm
  - Results will differ from one run to the other
  - The user's mental model might get challenged
- The scalability is not great

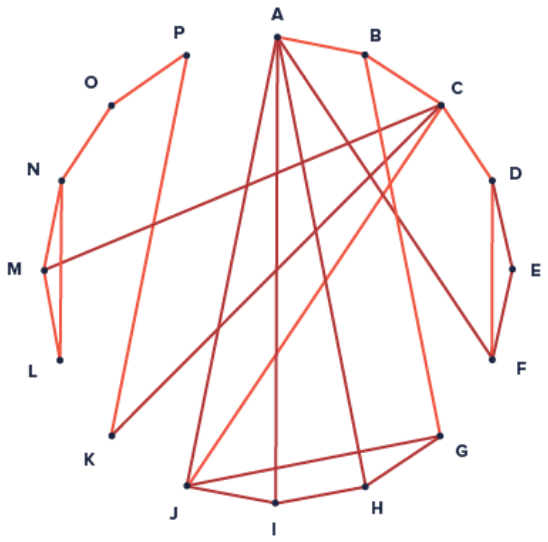
Chords



Chord Diagram



Non-ribbon Chord Diagram



Radial Network

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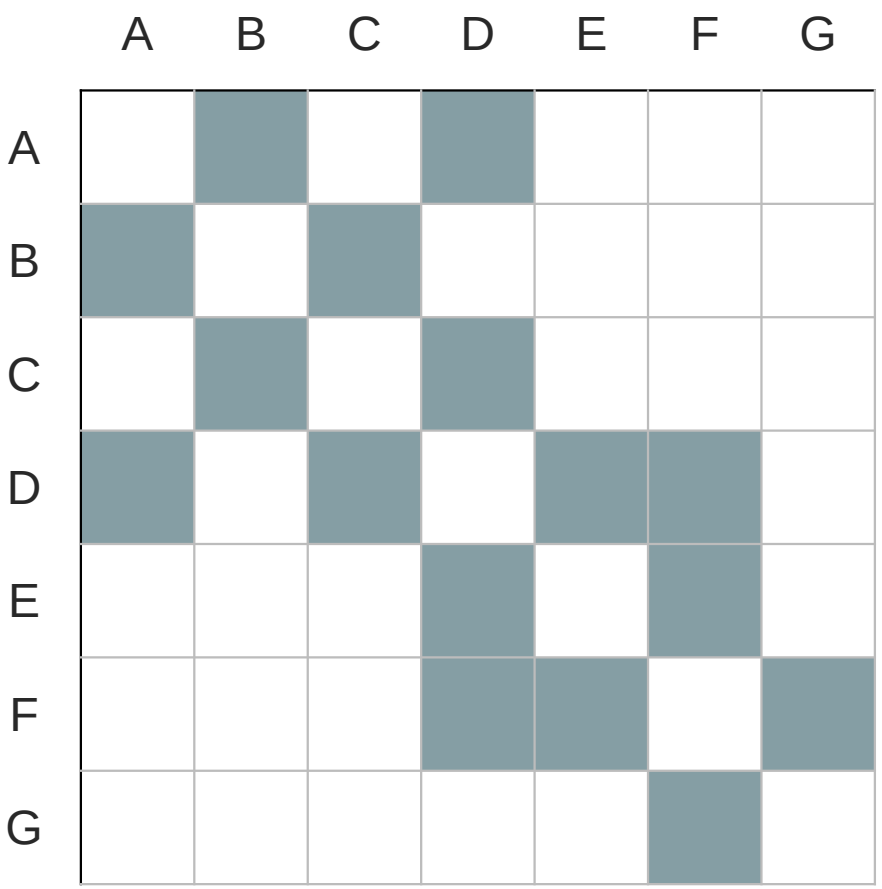
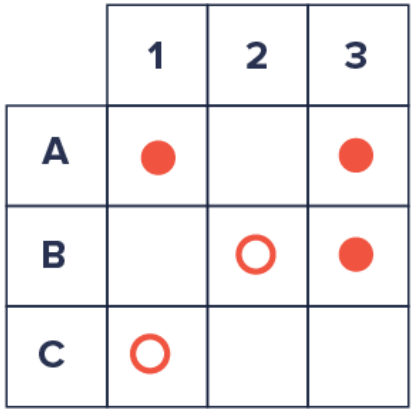
Adjacency

Arranging Nodes in Matrices

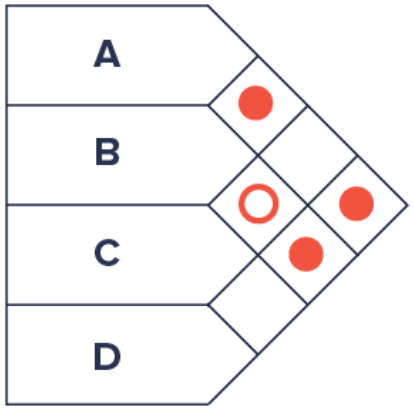
- Arrange the nodes in rows and columns
- Links become marks at the intersections of rows and columns
  - Channels like colour and shapes can be used to encode the links



Adjacency Matrices



Matrix Diagram



Roof-Shaped

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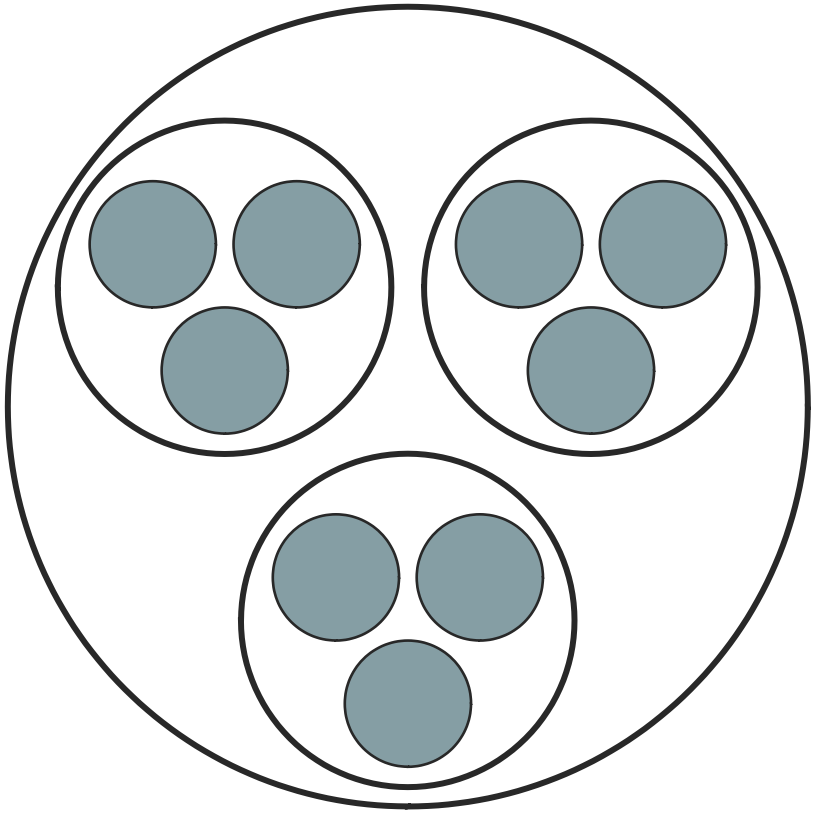
Node-Link vs. Adjacency Matrix

- Adjacency matrices manage to avoid the clutter seen on Node-Link diagrams
- They are also more predictable/stable and can be better reordered
- However, Node-Link diagrams are usually better to show the topology of a network
- Node-Link diagrams are also by far what users are familiar to

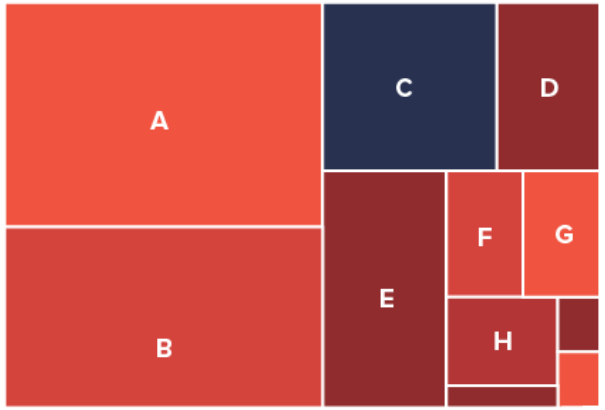
Enclosure

Enclosing Nodes

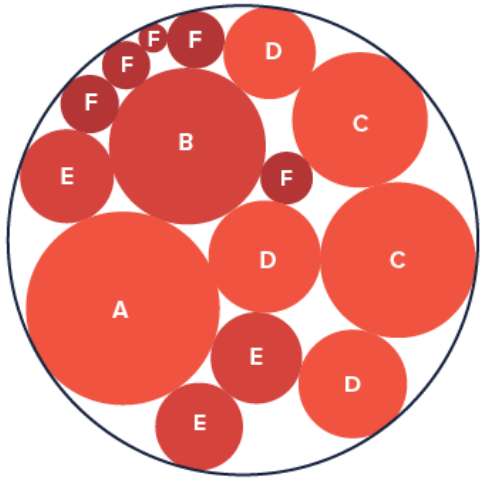
- This strategy only applies to trees
- Each node is an area
- Ancestor nodes contain their descendant
- This means that the size of a parent node depends on the size of its children



Enclosures



Treemap



Circle Pack

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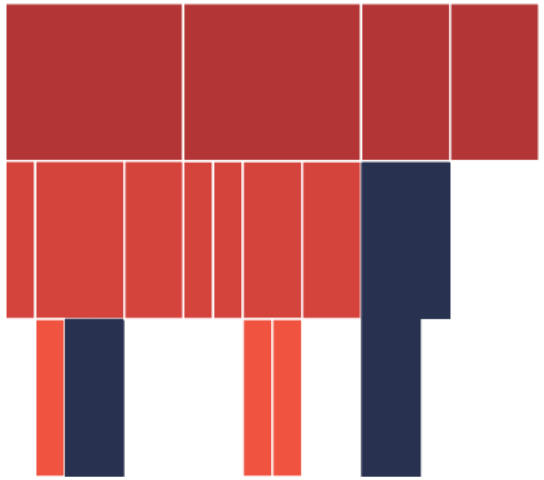
External Enclosure

Sunburst Diagram



Icicle Diagram

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Complex Networks

How many links can there be?

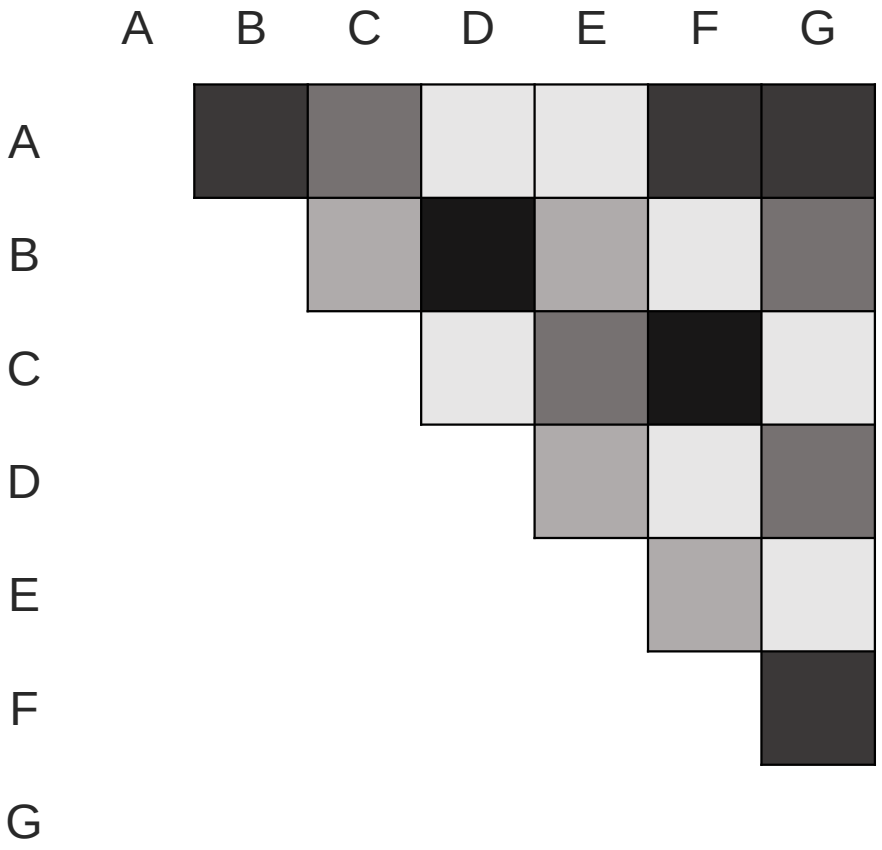
With a dataset of N nodes…

How many links can there be?

With a dataset of N nodes…

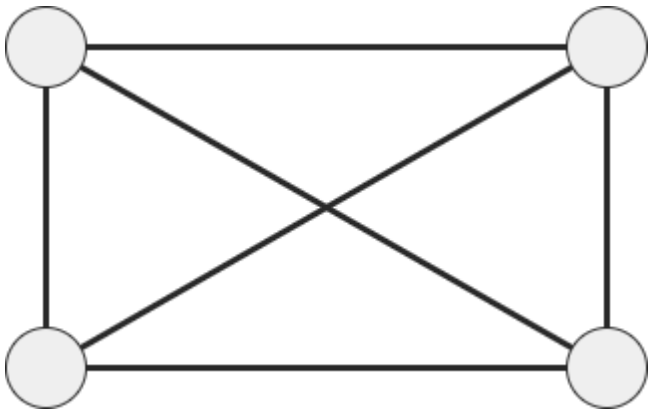
… there can be (N*(N-1)) / 2 links.

(if the space is metric)

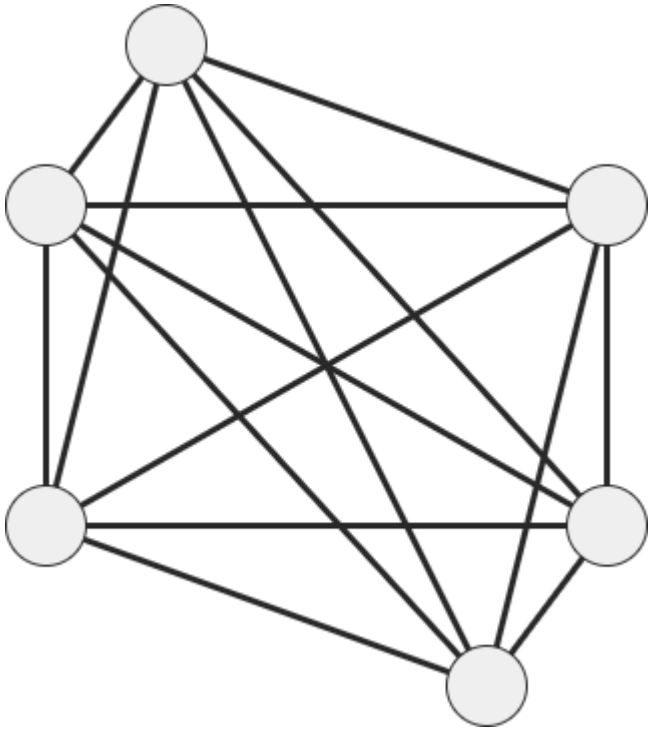


N=2

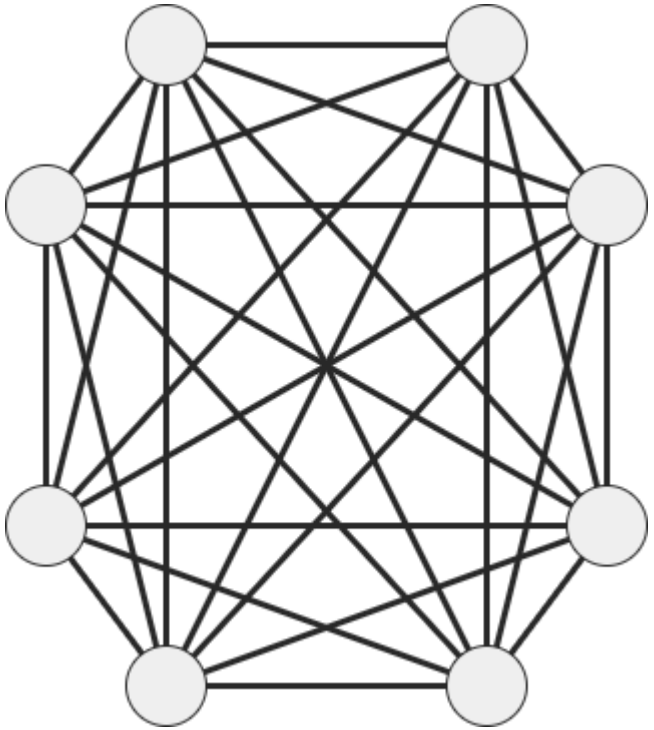
N=4



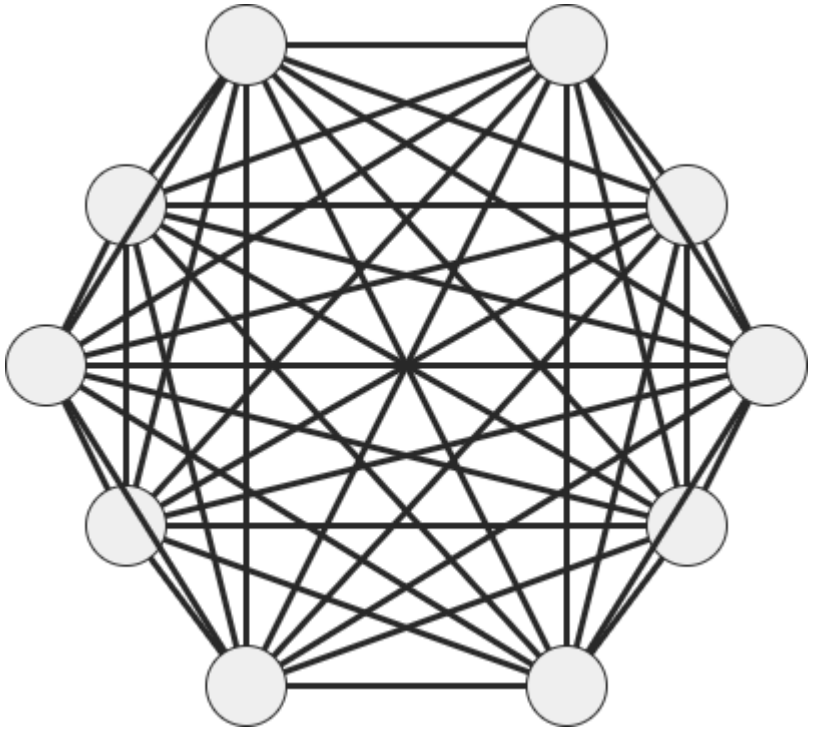
N=6



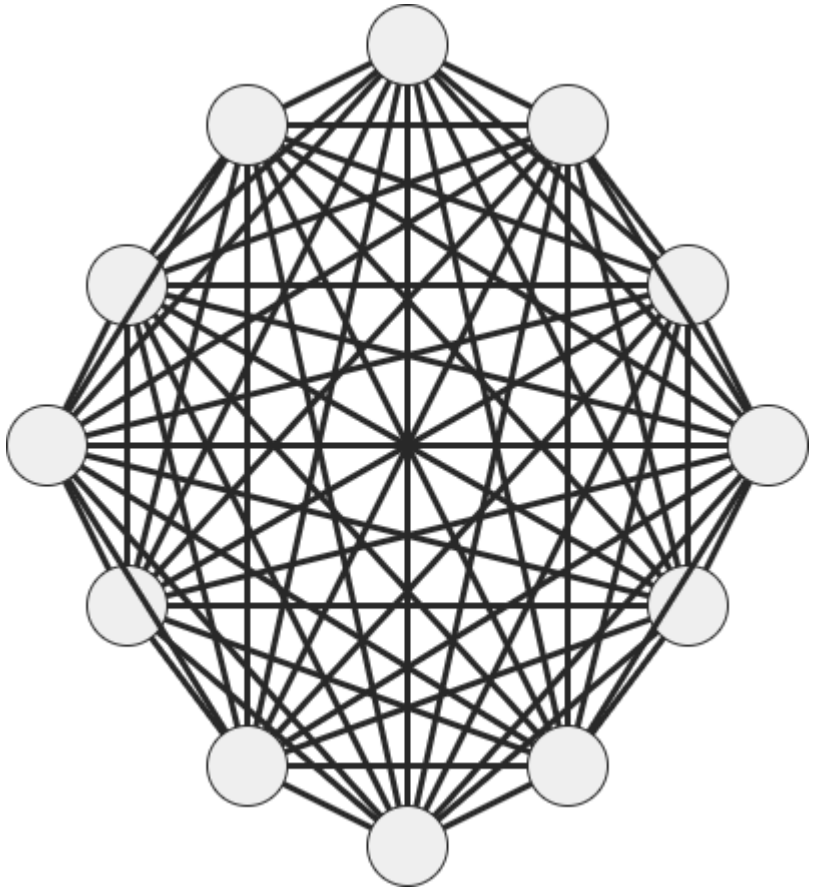
N=8



N=10

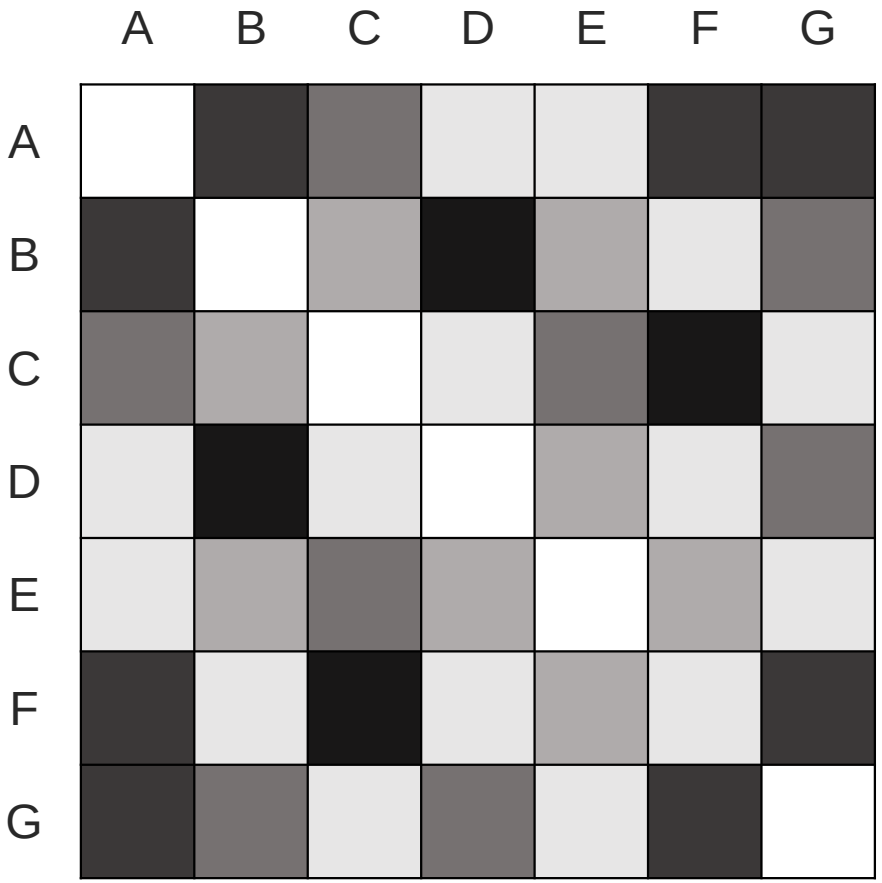


N=12



Dealing with Complex Networks

- Networks where all nodes are connected
- For example, similarity matrices



Solution 1

- Consider each row as an item
- Consider each column as a dimension
- Run a dimensionality reduction algorithm

key	1	2	3	4	5	6	7
A							
B							
C							
D							
E							
F							
G							

Solution 2

- Use hierarchical clustering to build a tree
- Each node becomes a leaf node
- Each parent node is an abstract cluster

Agglomerative Clustering

	A	B	C	D	E
A	1	0.2	0.6	0.1	0.3
B	0.2	1	0.3	0.7	0.5
C	0.6	0.3	1	0.2	0.4
D	0.1	0.7	0.2	1	0.4
E	0.3	0.5	0.4	0.4	1

Agglomerative Clustering

	A	B	C	D	E
A	1	0.2	0.6	0.1	0.3
B	0.2	1	0.3	**0.7**	0.5
C	0.6	0.3	1	0.2	0.4
D	0.1	**0.7**	0.2	1	0.4
E	0.3	0.5	0.4	0.4	1

- B, D, 0.7

Agglomerative Clustering

	A	B	C	D	E	**(B,D)**
A	1	0.2	0.6	0.1	0.3	**0.1**
B	0.2	1	0.3	**0.7**	0.5	
C	0.6	0.3	1	0.2	0.4	**0.2**
D	0.1	**0.7**	0.2	1	0.4	
E	0.3	0.5	0.4	0.4	1	**0.4**
(B,D)	**0.1**		**0.2**		**0.4**	**1**

- B, D, 0.7

Agglomerative Clustering

	A		C		E	**(B,D)**
A	1		0.6		0.3	**0.1**
C	0.6		1		0.4	**0.2**
E	0.3		0.4		1	**0.4**
(B,D)	**0.1**		**0.2**		**0.4**	**1**

- B, D, 0.7

Agglomerative Clustering

	A	C	E	**(B,D)**
A	1	0.6	0.3	**0.1**
C	0.6	1	0.4	**0.2**
E	0.3	0.4	1	**0.4**
(B,D)	**0.1**	**0.2**	**0.4**	**1**

- B, D, 0.7

Agglomerative Clustering

	E	(B,D)	**(A,C)**
E	1	0.4	**0.3**
(B,D)	0.4	1	**0.1**
(A,C)	**0.3**	**0.1**	**1**

- B, D, 0.7
- A, C, 0.6

Agglomerative Clustering

	(A,C)	**((B,D),E)**
(A,C)	1	**0.1**
((B,D),E)	**0.1**	**1**

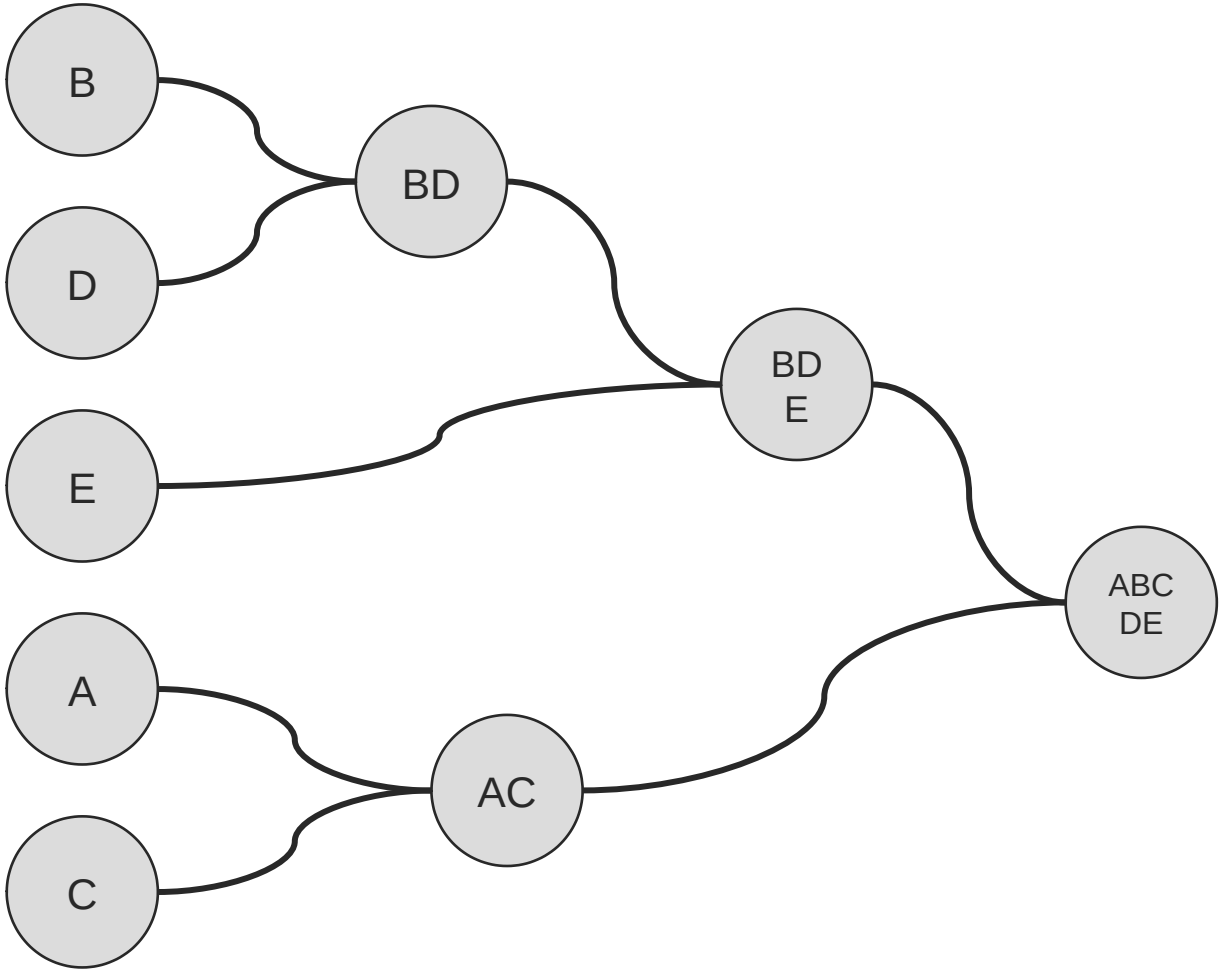
- B, D, 0.7
- A, C, 0.6
- (B,D), E, 0.4

Agglomerative Clustering

	(((B,D),E),(A,C))
(((B,D),E),(A,C))	1

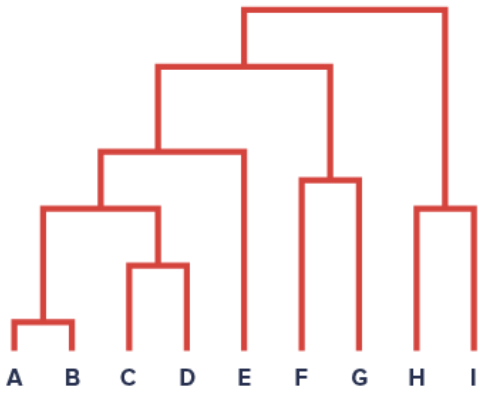
- B, D, 0.7
- A, C, 0.6
- (B,D), E, 0.4
- ((B,D),E), (A,C), 0.1

Agglomerative Clustering



- B, D, 0.7
- A, C, 0.6
- (B,D), E, 0.4
- ((B,D),E), (A,C), 0.1

Representing Hierarchical Clusters



Dendrogram

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Conclusion

Key Points

- **Networks** have the particularity to include both items and links
  - An item becomes a node or vertex
  - A link becomes an edge
  - Both can hold data attributes

- **Trees** are special types of network with a hierarchical semantic attached to the links

- To represent networks and trees:
  - **Connection**: drawing lines between nodes to represent edges
  - **Adjacency Matrices**: arranging nodes in a matrix, cells visualise links
  - **Enclosures**: for trees only, enclosing child-node marks in their parent-node marks